Name: _____

WORKSHEET

Scientific notation and significant figures

1 Write the following using scientific notation.

a 1 000 000 000 _____

b 230 _____

c 5 _____

d 0.000 1 _____

e 0.005 4 _____

f 0.7 _____

g 0.090 05 _____

h $\frac{6}{100000}$ _____

i 2^8 _____

j 4 006 100 _____

k 5000×10^2 _____

l 0.07×10^{-3} _____

2 How many significant figures do the following have?

a 502 000 _____

b 60 _____

c 7.990 0 _____

d 0.000 06 _____

e 0.500 30 _____

f 0.004 070 8 _____

g 300 401 _____

h 1 000.001 _____

i 782 _____

j 917.037 09 _____

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3 Write the following in scientific notation using three significant figures.

a 205 380 000 _____

b 186 500 _____

c 31.00005 _____

d 0.63190 _____

e 0.000 600 _____

f 0.050 067 09 _____

g 9 306 000 _____

h 9 325 000 _____

i 0.50 _____

j 89.177 812 _____

4 Complete the table.

Number	Number of significant figures	Scientific notation
7 300		
82 004 300		
21.09		
1.000 01		
0.003 400 020		
0.000 000 001		
0.100 01		
906.040 000		
1 000 001.001		

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Answers

1 **a** 1×10^{9}

b 2.3×10^{2}

c 5×10^{0}

d 1×10^{-4}

e 5.4×10^{-3}

f 7×10^{-1}

g 9.005×10^{-2}

h 6×10^{-5}

i 2.56×10^{2}

j 4.0061×10^{6}

k 5×10^{5}

l 7×10^{-5}

2 **a** 3

b 1

c 5

d 1

e 5

f 5

g 6

h 7

i 3

j 8

3 **a** 2.05×10^{8}

b 1.87×10^{5}

c 3.10×10^{1}

d 6.32×10^{-1}

e 6.00×10^{-4}

f 5.01×10^{-2}

g 9.31×10^{6}

h 9.33×10^{6}

i 5.00×10^{-1}

j 8.92×10^{1}

4

Number	Number of significant figures	Scientific notation
7 300	2	7.3×10^{3}
82 004 300	6	8.20043×10^{7}
21.09	4	2.109×10^{1}
1.000 01	6	1.00001×10^{0}
0.003 400 020	7	3.400020×10^{-3}
0.000 000 001	1	1×10^{-9}
0.100 01	5	1.0001×10^{-1}
906.040 000	9	9.06040000×10^{2}
1 000 001.001	10	$1.000001001 \times 10^{6}$

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